

SKILL WEEK 11

TASK-1: To Connect Modules, Validate Interfaces, Test End-to-End Execution, Handle Errors, Improve Reliability, Refactor 4 Components

Ans:

```
arikel@DESKTOP-AMUSJME:~/OSSP$ gcc -Wall -Wextra -g -Iinclude src/*.* -o mysh
arikel@DESKTOP-AMUSJME:~/OSSP$ ./mysh
myShell> echo Hello

--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello
Child PID: 794
Hello
Command execution completed.
myShell> pwd

--- Parsed Arguments ---
Argument 1: pwd
Child PID: 795
/home/arikel/OSSP
Command execution completed.
myShell> ls

--- Parsed Arguments ---
Argument 1: ls
Child PID: 796
Makefile  assets  include  scripts  task1.c  tests
OSSP      bin     mysh     src       task2
README.md docs    obj      task1     task2.c
Command execution completed.
myShell> exit
Exiting shell.
arikel@DESKTOP-AMUSJME:~/OSSP$
```

TASK-2: TO Detect Runtime Errors, Generate Messages, Log Failures, Handle Invalid Syntax, Recover Gracefully, Test Error Scenarios

Ans:

```
arike1a@DESKTOP-AMUSJME:~/OSSP$ ./mysh
myShell> abcdef

--- Parsed Arguments ---
Argument 1: abcdef
Child PID: 833
Command execution failed: No such file or directory
Command execution completed.
myShell> cat nonexistent.txt

--- Parsed Arguments ---
Argument 1: cat
Argument 2: nonexistent.txt
Child PID: 836
cat: nonexistent.txt: No such file or directory
Command execution completed.
myShell> cd /does/not/exist

--- Parsed Arguments ---
Argument 1: cd
Argument 2: /does/not/exist
Child PID: 837
Command execution failed: No such file or directory
Command execution completed.
myShell> echo Still working

--- Parsed Arguments ---
Argument 1: echo
Argument 2: Still
Argument 3: working
Child PID: 838
Still working
Command execution completed.
```

TASK-3: To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings

```
arikel@DESKTOP-AMUSJME: x + v - □ ×
arikel@DESKTOP-AMUSJME:~/OSSP$ valgrind --leak-check=full --show-leak-kinds=all ./mysh
==869== Memcheck, a memory error detector
==869== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==869== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==869== Command: ./mysh
==869==
myShell> echo Hello

--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello
Child PID: 872
Hello
Command execution completed.
myShell> pwd

--- Parsed Arguments ---
Argument 1: pwd
Child PID: 875
/home/arikel/OSSP
Command execution completed.
myShell> exit
Exiting shell.
==869==
==869== HEAP SUMMARY:
==869==   in use at exit: 0 bytes in 0 blocks
==869== total heap usage: 2 allocs, 2 frees, 2,048 bytes allocated
==869==
==869== All heap blocks were freed -- no leaks are possible
==869==
==869== For lists of detected and suppressed errors, rerun with:
==869== -s
==869== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 f
```

TASK-4: To Design Test Cases, Create Test Scripts, Automate Execution, Verify Outputs, Compare Results, Generate Reports

Ans:

```
arikel@DESKTOP-AMUSJME: × + ▾
GNU nano 8.7.1 test_shell.sh *
#!/bin/bash

echo "==== TEST 1: echo ====="
echo "echo Hello" | ./mysh

echo "==== TEST 2: pwd ====="
echo "pwd" | ./mysh

echo "==== TEST 3: invalid command ====="
echo "abcdef" | ./mysh

echo "==== TEST 4: multiple commands ====="
printf "echo First\npwd\nnecho Second\nexit\n" | ./mysh

echo "==== ALL TESTS COMPLETED ====="
```

```
arikel@DESKTOP-AMUSJME: x + v - □
arikel@DESKTOP-AMUSJME:~/OSSP$ cat tests/tests_parser.c
arikel@DESKTOP-AMUSJME:~/OSSP$ cat tests/test_executor.c
arikel@DESKTOP-AMUSJME:~/OSSP$ nano test_shell.sh
arikel@DESKTOP-AMUSJME:~/OSSP$ chmod +x test_shell.sh
arikel@DESKTOP-AMUSJME:~/OSSP$ ./test_shell.sh
===== TEST 1: echo =====
myShell>
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello
Child PID: 995
Hello
Command execution completed.
myShell> ===== TEST 2: pwd =====
myShell>
--- Parsed Arguments ---
Argument 1: pwd
Child PID: 998
/home/arikel/OSSP
Command execution completed.
myShell> ===== TEST 3: invalid command =====
myShell>
--- Parsed Arguments ---
Argument 1: abcdef
Child PID: 1001
Command execution failed: No such file or directory
Command execution completed.
myShell> ===== TEST 4: multiple commands =====
myShell>
--- Parsed Arguments ---
Argument 1: echo
Argument 2: First
Child PID: 1004
First
Command execution completed.
myShell>
--- Parsed Arguments ---
```

```
arikel@DESKTOP-AMUSJME: x + v - □ X
--- Parsed Arguments ---
Argument 1: pwd
Child PID: 1005
/home/arikel/OSSP
Command execution completed.
myShell>
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Second
Child PID: 1006
Second
Command execution completed.
myShell> Exiting shell.
===== ALL TESTS COMPLETED =====
arikel@DESKTOP-AMUSJME:~/OSSP$ ./test_shell.sh > test_results.t
xt 2>&1
```